

CE 206: Engineering Computation Sessional

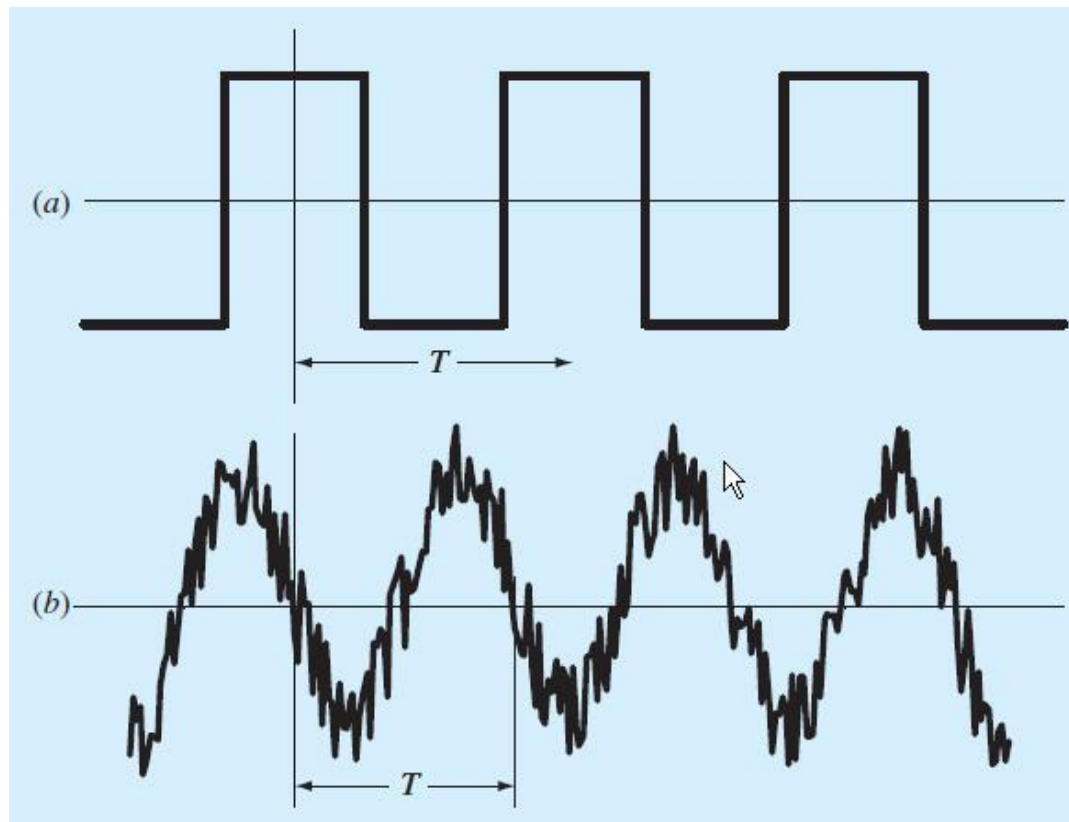
Fourier Analysis

Periodic Functions

A periodic function is one for which

$$f(t) = f(t + T)$$

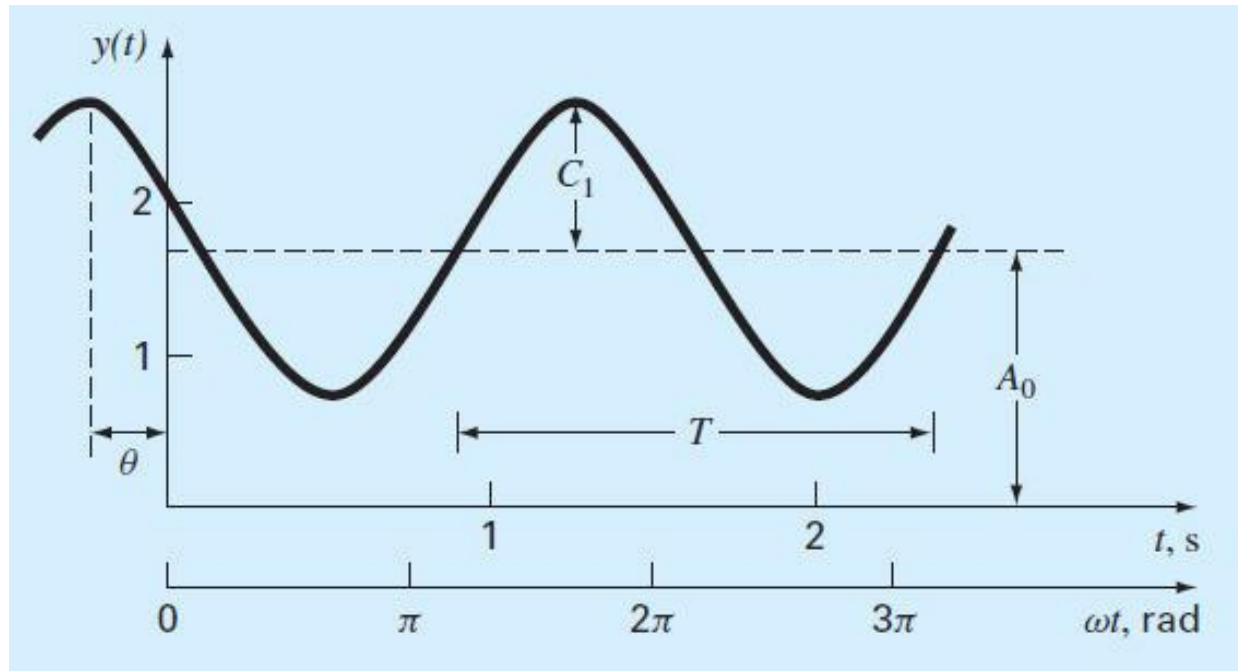
where T = the period



Sinusoids

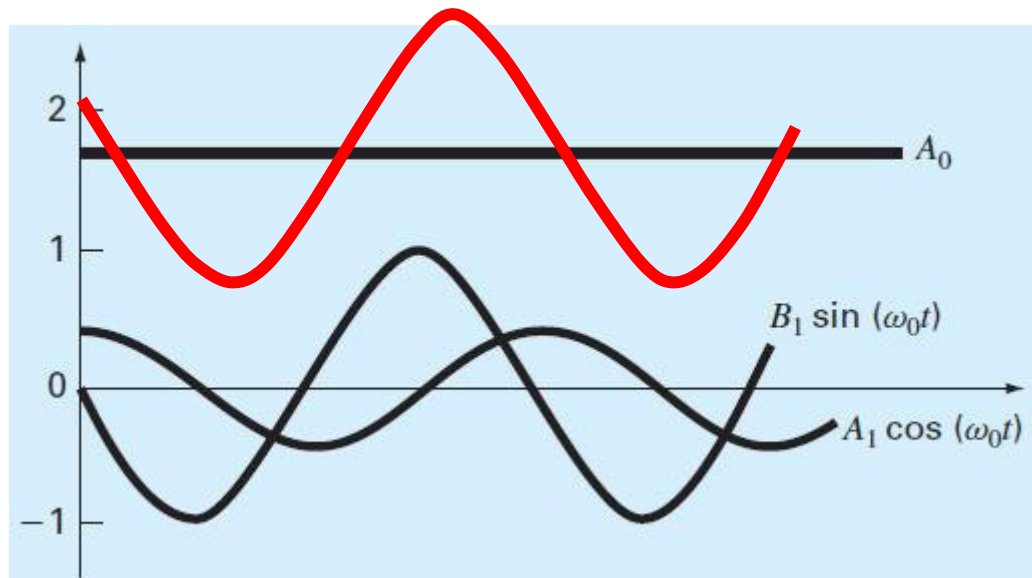
$$f(t) = A_0 + C_1 \cos(\omega_0 t + \theta)$$

Mean value Amplitude Angular frequency Phase shift



Alternative Expressions of Sinusoids

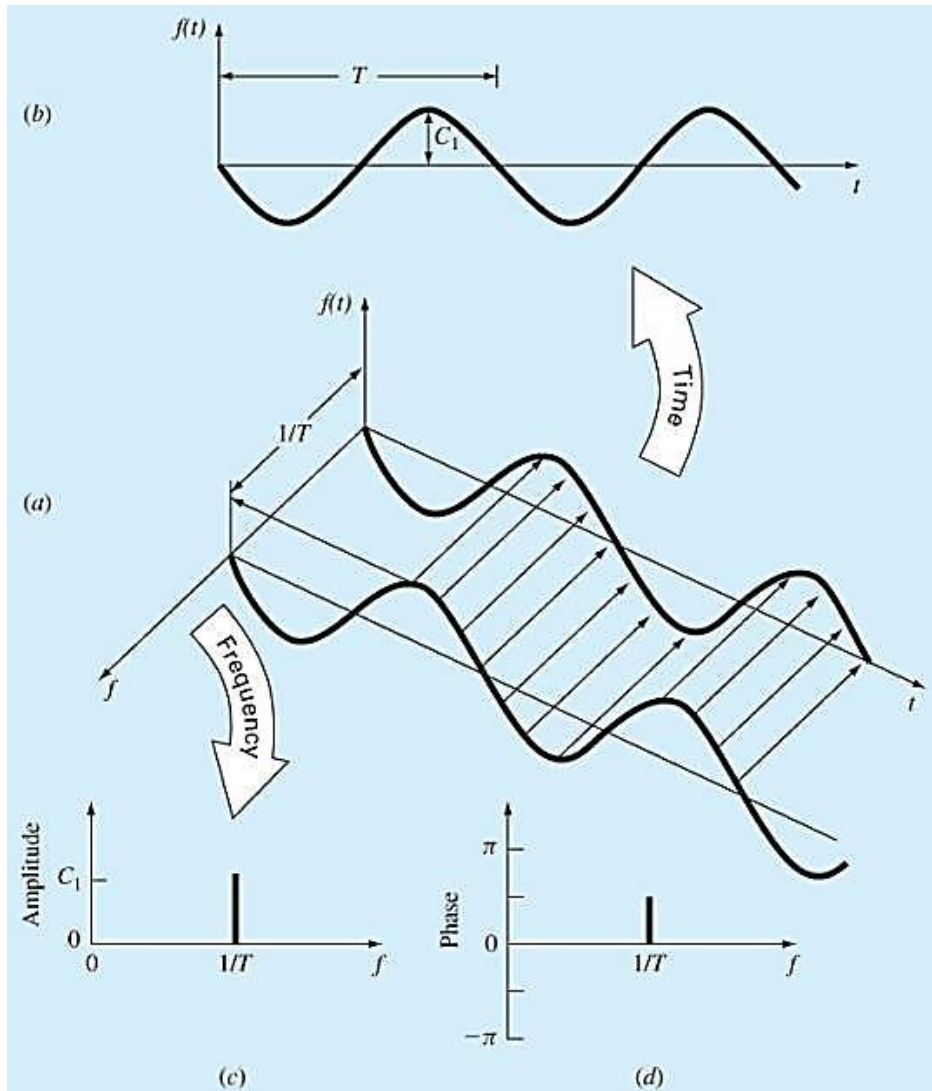
$$f(t) = A_0 + A_1 \cos(\omega_0 t) + B_1 \sin(\omega_0 t)$$



The two forms are related by

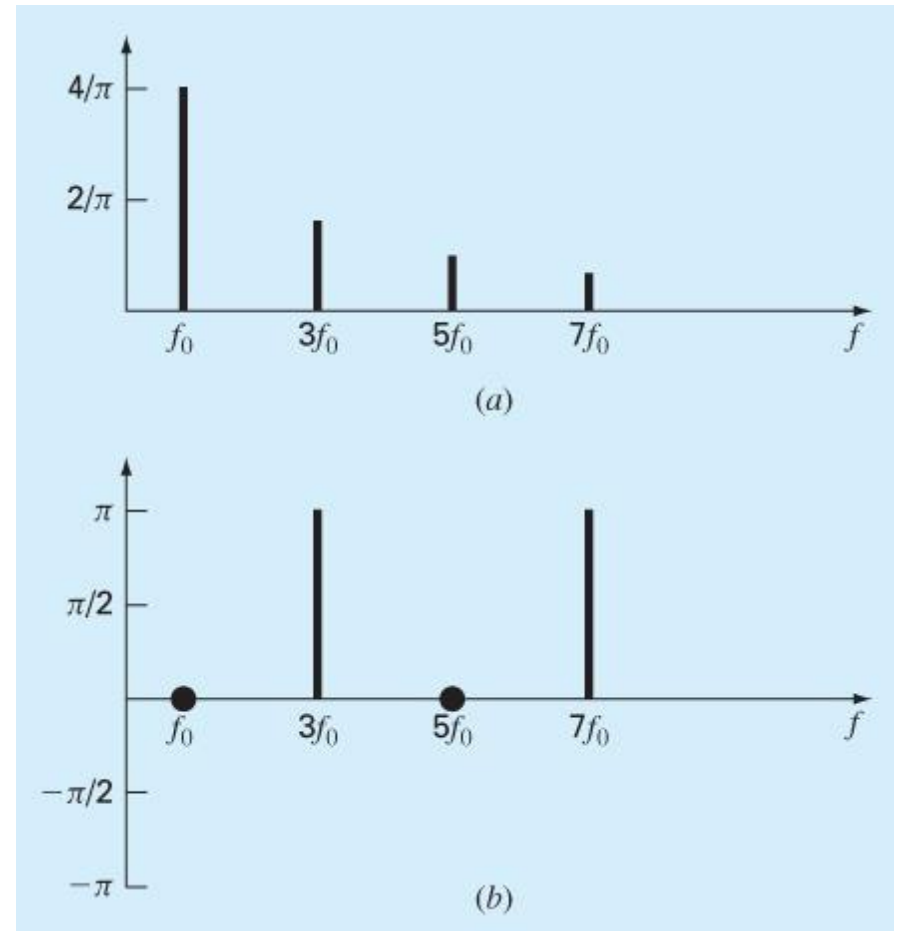
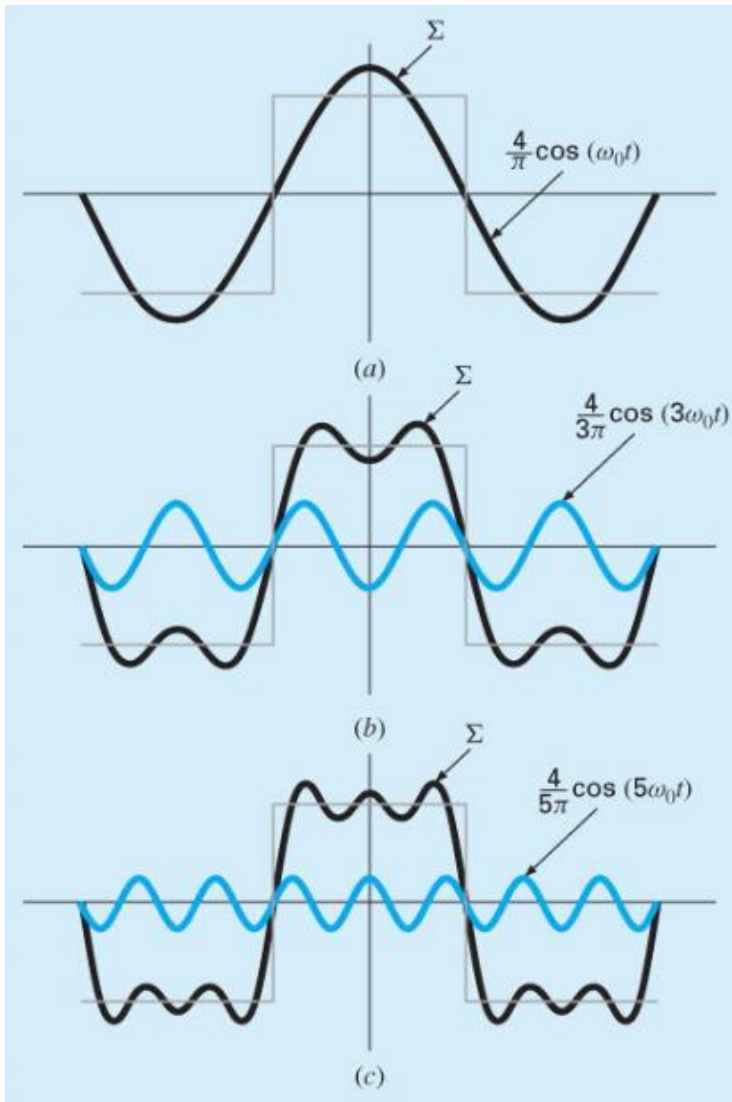
$$C_1 = \sqrt{A_1^2 + B_1^2} \qquad \theta = \arctan(-B_1/A_1)$$

Time vs. Frequency Domains



Frequency domain provides an alternative perspective for characterizing the behavior of oscillating functions

Fourier Series approx. and transform



Fourier Series and Fourier Transform

Continuous Fourier Series:

$$f(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)]$$

Coefficients:

$$a_k = \frac{2}{T} \int_0^T f(t) \cos(k\omega_0 t) dt$$

$$b_k = \frac{2}{T} \int_0^T f(t) \sin(k\omega_0 t) dt$$

Compact form:

$$f(t) = \sum_{k=-\infty}^{\infty} \tilde{c}_k e^{ik\omega_0 t}$$

Coefficients:

$$\tilde{c}_k = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-ik\omega_0 t} dt$$

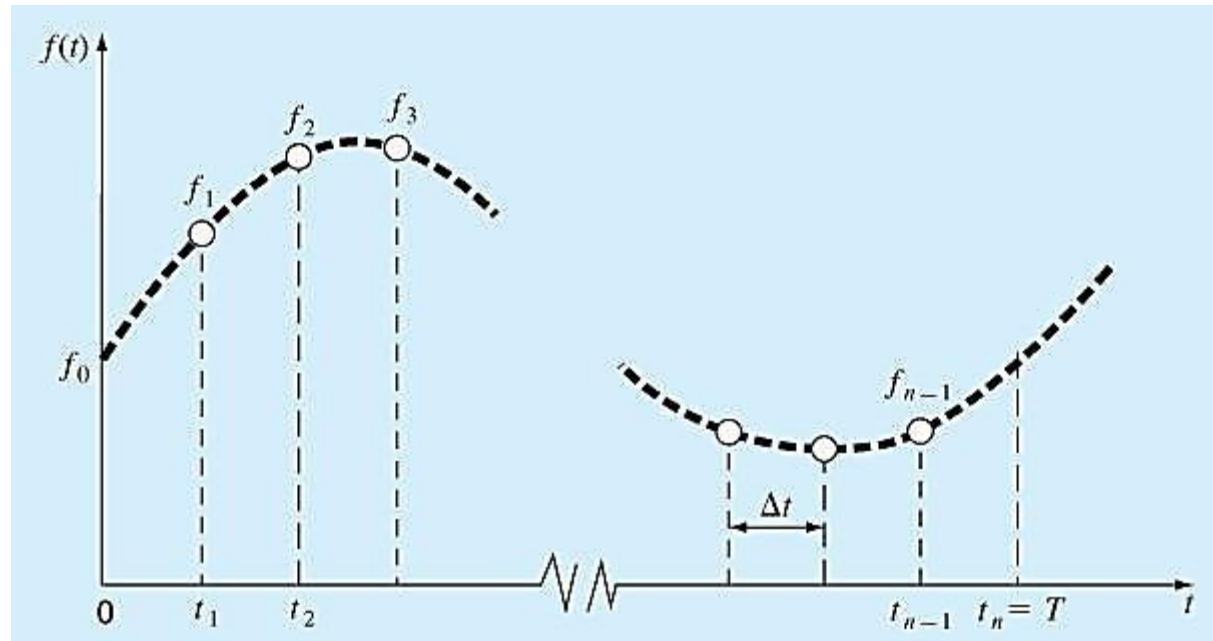
Discrete Fourier Transform (DFT)

In engineering, data are often collected in or converted to discrete format.

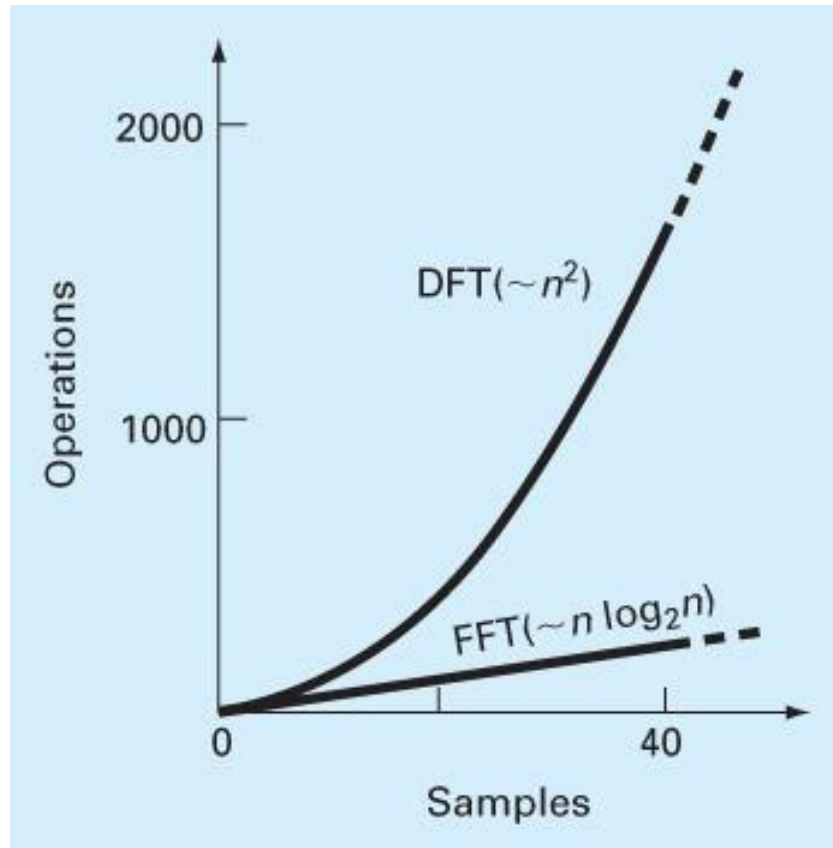
Discrete Fourier transform: $F_k = \sum_{j=0}^{n-1} f_j e^{-ik\omega_0 j}$ for $k = 0$ to $n-1$

Inverse Fourier transform: $f_k = \frac{1}{n} \sum_{k=0}^{n-1} F_k e^{ik\omega_0 j}$ for $j = 0$ to $n-1$

$$\omega_0 = \frac{2\pi}{n}$$



MATLAB built-in function: `fft`



DFT is computationally burdensome.

A Fast Fourier Transform (FFT) algorithm has been developed to compute the DFT in an economical fashion.

MATLAB has a built-in function to compute FFT

$$\mathbf{F} = \text{fft}(\mathbf{f}, n)$$

$\mathbf{F}$ = a vector containing the DFT

$\mathbf{f}$ = a vector containing the signal/ discrete measurements

Example Problem: DFT

Apply MATLAB's fft function to determine the DFT for a simple sinusoid:

$$f(t) = 5 + \cos(2\pi(12.5)t) + \sin(2\pi(18.75)t)$$

Generate 8 equispaced points with $\Delta t = 0.02$ s. Plot the results versus frequency

Solution:

Sampling frequency = $(1/0.02\text{s}) = 50\text{Hz}$

Total sample length = $8/50 = 0.16\text{s}$

The Nyquist frequency = $0.5 \cdot 50 = 25\text{Hz}$

(The highest frequency that can be measured in a signal is called the **Nyquist frequency** and is half the sampling frequency)

Lowest detectable frequency = $1/0.16\text{s} = 6.25\text{Hz}$

Therefore, the analysis should be able to detect signals with periods from $1/25 = 0.04$ upto $1/6.25 = 0.16\text{s}$.

Example Problem: DFT

```
clc; clear all; close all;
n=8; dt=0.02; fs=1/dt; T=0.16;
tspan=(0:n-1)/fs;
y=5+cos(2*pi*12.5*tspan)+sin(2*pi*18.75*tspan);
subplot(3,1,1);
plot(tspan,y,'-
ok','linewidth',2,'MarkerFaceColor','black');
title('(a) f(t) versus time (s)');
%%%%%%%%%%
Y=fft(y)/n;
Y'
```

```
ans =
    5.0000
    0.0000 - 0.0000i
    0.5000
   -0.0000 + 0.5000i
         0
   -0.0000 - 0.5000i
    0.5000
    0.0000 + 0.0000i
```

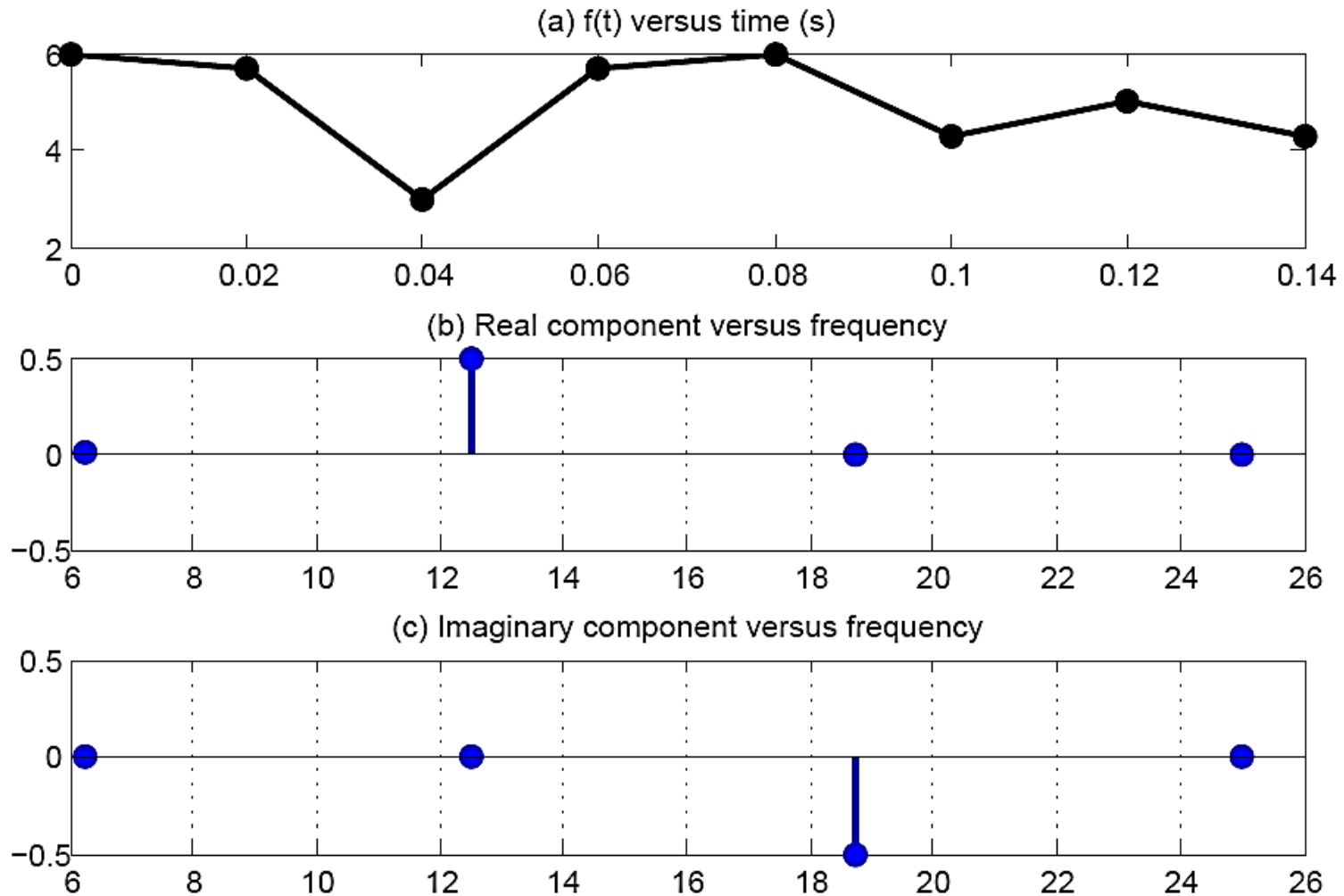
Example Problem: DFT

```
figure(1)
nyquist=fs/2; fmin=1/T;
f=linspace(fmin,nyquist,n/2);
Y(1)=[]; YP=Y(1:n/2);

subplot(3,1,2);
stem(f,real(YP),'linewidth',2,'MarkerFaceColor','blue');
grid; title('(b) Real component versus frequency')

subplot(3,1,3);
stem(f,imag(YP),'linewidth',2,'MarkerFaceColor','blue');
grid; title('(c) Imaginary component versus frequency')
```

Example Problem: DFT



Power Spectrum

Provides another useful way to discern the underlying harmonics of seemingly random signals.

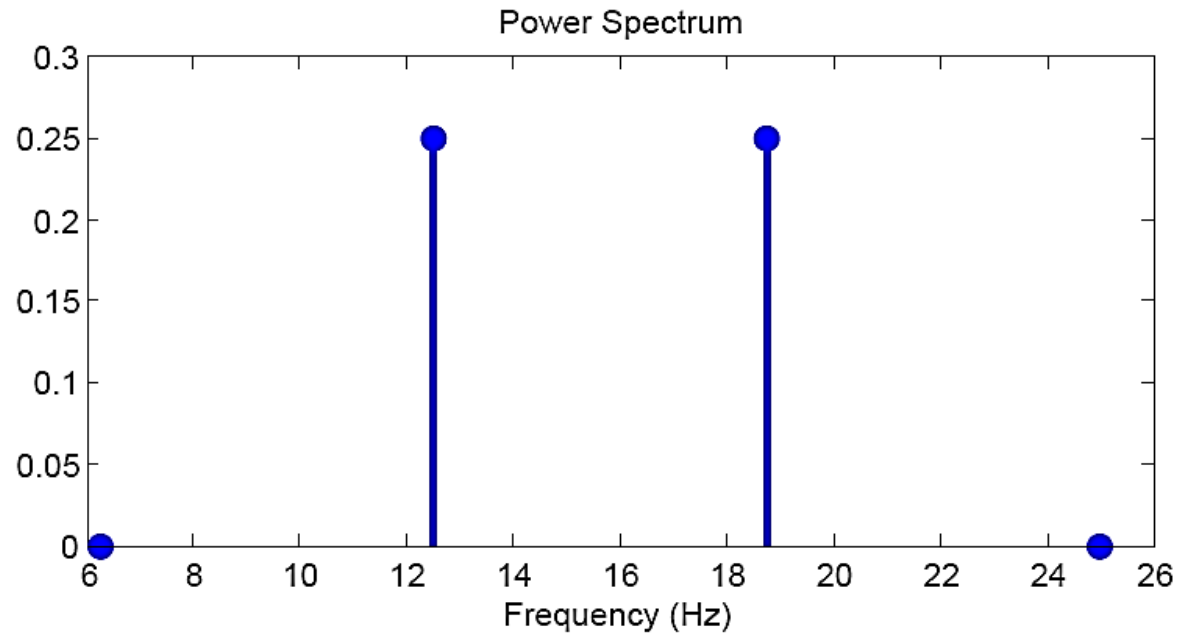
In terms of DFT, a power spectrum consists of a plot of power associated with each frequency component (P_k) versus frequency.

Power = sum of
square of Fourier
coefficients

$$P_k = |\tilde{c}_k|^2$$

figure (2)

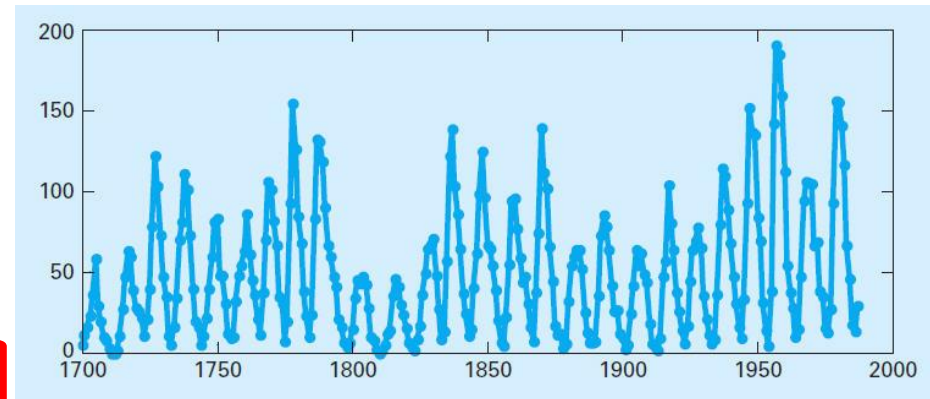
```
Pyy=abs(YP).^2;  
stem(f,Pyy,'linewidth',2,'MarkerFaceColor','blue');  
title('Power Spectrum'); xlabel('Frequency (Hz)');
```



Example Problem: Wolf Sunspot Number

The data for year-wise sunspot number are contained in the MATLAB file **sunspot.dat**. Perform Fourier analysis on the time series data and determine the cycle length. Compare it to J.R. Wolf's estimate of cycle length (~ 11.1 years).

```
clear all; close all;  
load sunspot.dat  
year=sunspot(:,1);  
number=sunspot(:,2);  
  
n=length(number);  
a=polyfit(year, number,1);  
lineartrend=polyval(a,year);  
ft=number-lineartrend;  
  
F=fft(ft);  
fs=1; f=(0:n/2)*fs/n;  
pow=abs(F(1:n/2+1)).^2;
```



Removing the
upward linear
trend of the data

Example Problem: Wolf Sunspot Number

```
plot(f,pow)
xlabel('Frequency (cycles/year)'); ylabel('power')
title('Power versus frequency')
```

